

ALyl Chem 17 EQ P2 22w to 09s Paper 2 Carbonyl compounds 75marks

As you start and work through this worksheet you can tick off your progress to show yourself how much you have done, and what you need to do next. The first task is just to read the first question and should take you less than one minutes to complete.

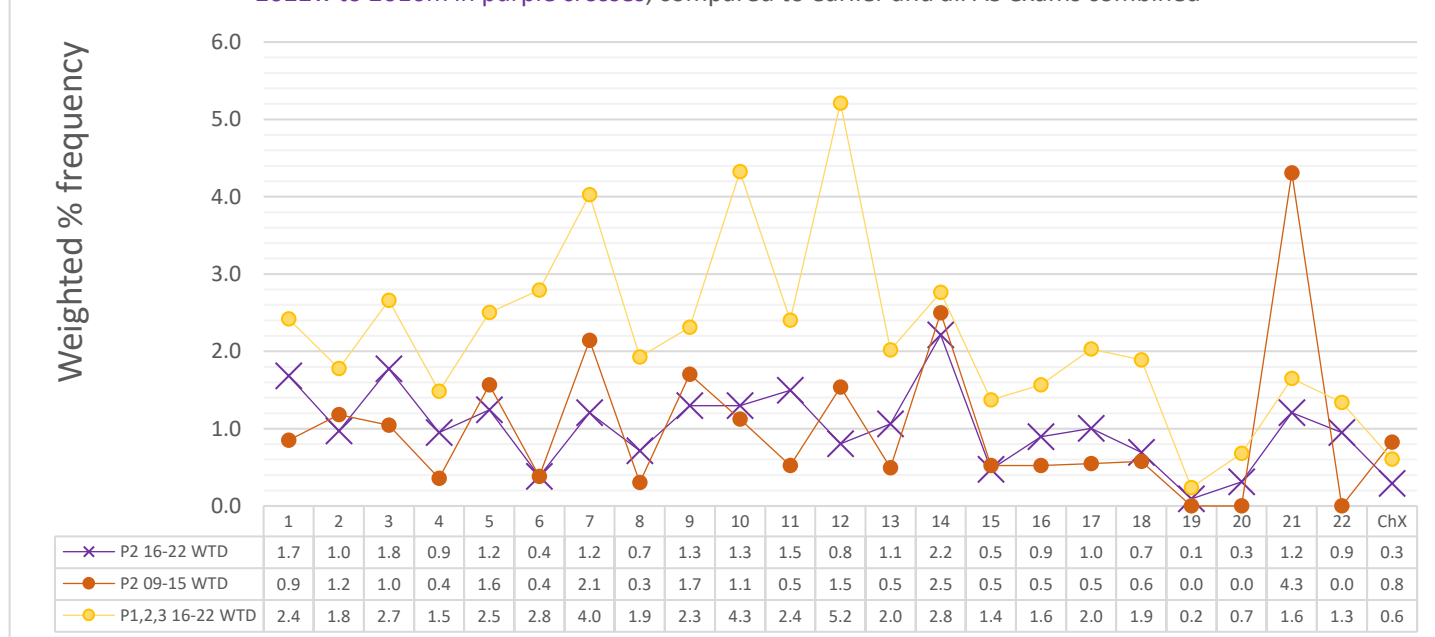
Paper 2 Topic 17

Checklist Tick each task off as you go along

RANK:

P1 Noob	P1 Novice	P1 Bronze	P1 Silver	P1 Gold	P1 ¹ Winner	P1 Hero	P1 Legend
1 Q started	1 Q done	10% of marks	25% of marks	40% of marks	50% of marks	75% of marks	100% of marks
	4	8	19	30	38	56	75
	5	9	23	38	47	70	94

9701 Chemistry Weighted Mark Frequency: **Paper 2**
2022w to 2016m in purple crosses, compared to earlier and all AS exams combined



What the most thoughtful students will get out of their extensive studying will be a capacity to do meaningful brain-based work even under stressful conditions, which is a part of the self-mastery skillset that will continue to deliver value for the whole of their lives. Outstanding grades will also happen, but the most important goal from skillful action in study is being better at any important task, even if circumstances do not feel ideal.

As you are moving through your studies you can learn more about yourself by trying out new ways to manage yourself, and analysing how effective those new techniques were. In this reflective process not only will you get better at working positively and productively to deliver ambitious and successful outcomes, but you will be working towards one aspect of life's highest pursuit, summarised and inscribed on the Temple of Apollo at Delphi: "know thyself".

1. To complete these questions, as important as your answer, is checking your answer against the mark scheme.
2. For each page or group of 10-20 marks, convert your mark score into a percentage. This will allow you to see (and feel) your progress as you get more experience and understanding with each topic.
3. Multiple choice questions, done carefully where you explain and show yourself your thinking using written notes as you move through each question, can be more useful than just Paper 2 for students aiming for a C or B grade. Paper 2 should be the larger focus for students aiming for A and A* grades, however.
4. If you find you get a higher percentage answering short answer questions than multiple choice questions that often means you are NOT using the marking scheme correctly; your correct answer might not be fully complete for all the marks you are awarding. The marks easiest to miss rely on providing the largest amount of detail.

¹ **DO NOT** work on these higher levels of completion in your AS year unless you have also achieved at least a "Silver" (25%) in the same topic in **Paper 1**, which tend also to be easier questions, as well as "Silver" (25%) in the same topic, if it exists, in Paper 3.

17 Carbonyl compounds

17.1 Aldehydes and ketones

Learning outcomes

Candidates should be able to:

- recall the reactions (reagents and conditions) by which aldehydes and ketones can be produced:
 - the oxidation of primary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce aldehydes
 - the oxidation of secondary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce ketones
- describe:
 - the reduction of aldehydes and ketones, using NaBH_4 or LiAlH_4 to produce alcohols
 - the reaction of aldehydes and ketones with HCN , KCN as catalyst, and heat to produce hydroxynitriles exemplified by ethanal and propanone
- describe the mechanism of the nucleophilic addition reactions of hydrogen cyanide with aldehydes and ketones in 17.1.2(b)
- describe the use of 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) to detect the presence of carbonyl compounds
- deduce the nature (aldehyde or ketone) of an unknown carbonyl compound from the results of simple tests (Fehling's and Tollens' reagents; ease of oxidation)
- deduce the presence of a $\text{CH}_3\text{CO}-$ group in an aldehyde or ketone, $\text{CH}_3\text{CO}-\text{R}$, from its reaction with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate of tri-iodomethane and an ion, RCO_2^-

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(c) **W**, **X** and **Y** have the same molecular formula, $\text{C}_5\text{H}_{10}\text{O}$.

W, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.

Table 4.1

	+ 2,4-dinitrophenylhydrazine	+ alkaline $\text{I}_2(\text{aq})$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(i) **W**, **X** and **Y** each contain a common functional group.

Name the functional group that is present in all three compounds.

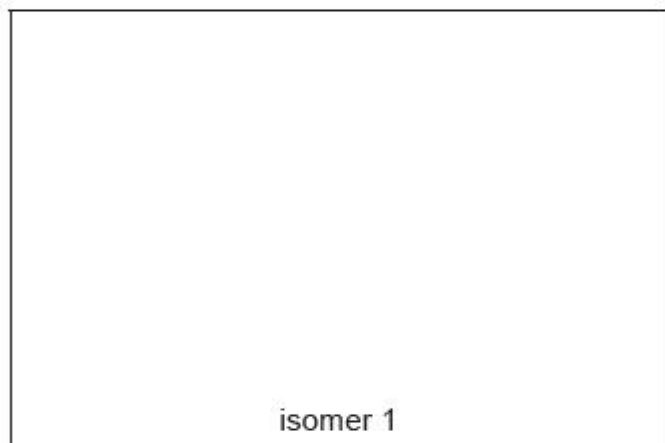
..... [1]

(ii) State the formula of the yellow precipitate produced when **X** is added to alkaline $\text{I}_2(\text{aq})$.

..... [1]

(iii) **W** could be one of four structural isomers.

- Draw the skeletal formulae for two possible structural isomers of **W**.
- Describe the type of structural isomerism shown.



type of structural isomerism

[3]

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4 Compounds **J** and **K** are found in plant oils.

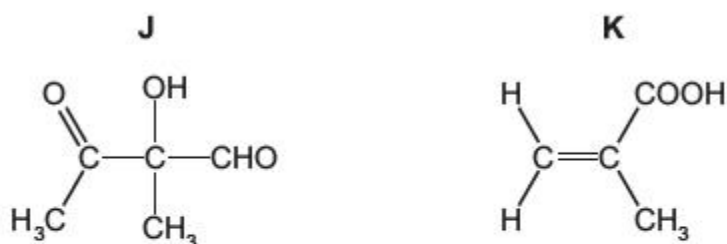


Fig. 4.1

(a) (i) Complete Table 4.1 to state what you would **observe** when **J** reacts with the reagents listed.

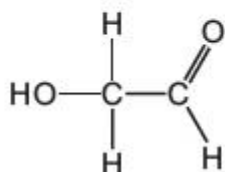
Table 4.1

reagent	observation with J
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Tollens' reagent	
sodium metal	

[3]

4 Hydroxyethanal, HOCH_2CHO , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



- (b) Hydroxyethanal reacts separately with 2,4-dinitrophenylhydrazine (2,4-DNPH) and with Tollens' reagent.

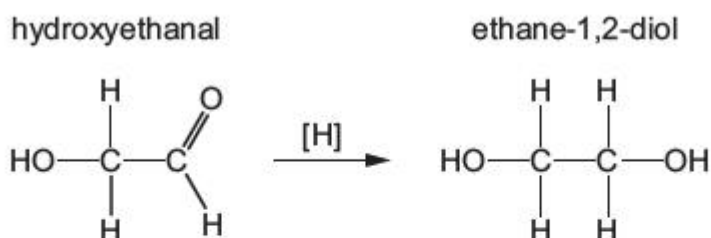
State what you would observe in each reaction.

reaction with 2,4-DNPH

reaction with Tollens' reagent

[2]

- (d) Hydroxyethanal can be reduced to ethane-1,2-diol, $(\text{CH}_2\text{OH})_2$, as shown.



- (i) Write an equation for the reduction of hydroxyethanal to $(\text{CH}_2\text{OH})_2$.

Use $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

..... [1]

- (ii) Identify a reagent for this reduction reaction.

..... [1]

4 Iodine is used in many inorganic and organic reactions.

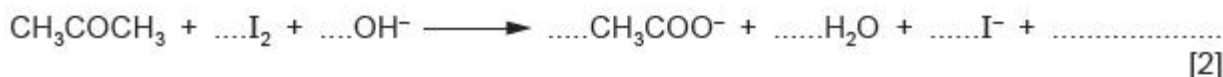
Iodoalkanes contain carbon-iodine bonds.

The simplest iodoalkane is CH_3I .

- (i) CH_3I can be made from methanol, CH_3OH .

- (vi) Propanone reacts with excess alkaline aqueous iodine.

Complete and balance the equation for this reaction.



[2]

- (vii) State **one** observation that can be made in the reaction in (c)(vi).

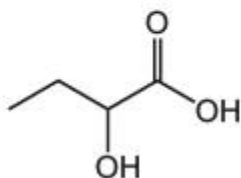
..... [1]

3 The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

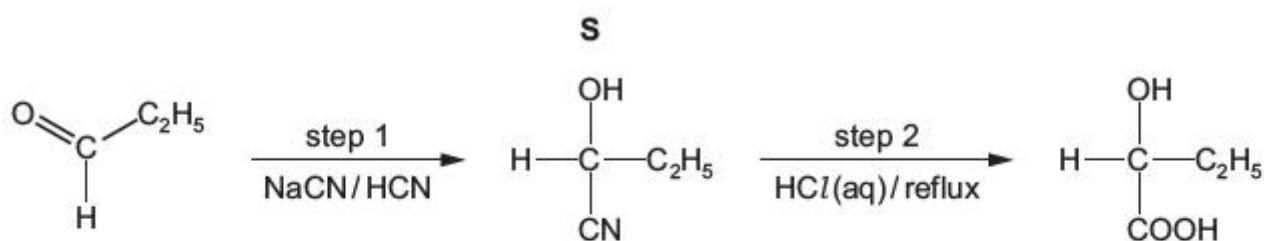
(a) (i) At 200°C , aluminium chloride exists as $\text{Al}_2\text{Cl}_6(\text{g})$.

(c) Two students try to prepare 2-hydroxybutanoic acid in the laboratory.

2-hydroxybutanoic acid



A third student prepares 2-hydroxybutanoic acid using propanal as the starting material. In step 1 the student reacts propanal with a mixture of NaCN and HCN .



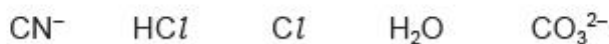
(iii) Draw the mechanism for the reaction of propanal with the mixture of NaCN and HCN to form **S**.

- Identify the ion that reacts with propanal.
- Draw the structure of the intermediate of the reaction.
- Include all charges, partial charges, lone pairs and curly arrows.



[4]

5 (a) Below is a list of species which can react with organic compounds.



(c) (i) Name an organic functional group which reacts with a nucleophile in an addition reaction.

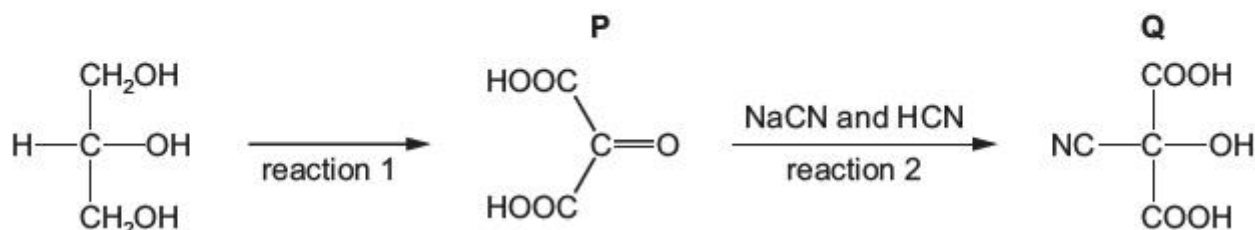
..... [1]

(ii) Name an organic functional group which tends to react with a nucleophile in an $\text{S}_{\text{N}}1$ substitution mechanism.

..... [1]

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(a) A series of reactions starting from glycerol is shown.



(ii) Name the reaction mechanism for reaction 2.

..... [1]

(iii) Give the observation you would make when 2,4-dinitrophenylhydrazine is added to P.

..... [1]

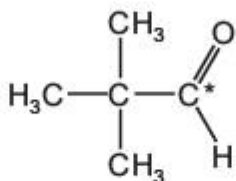
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4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(a) (i) Give the name of $(\text{CH}_3)_3\text{CCHO}$.

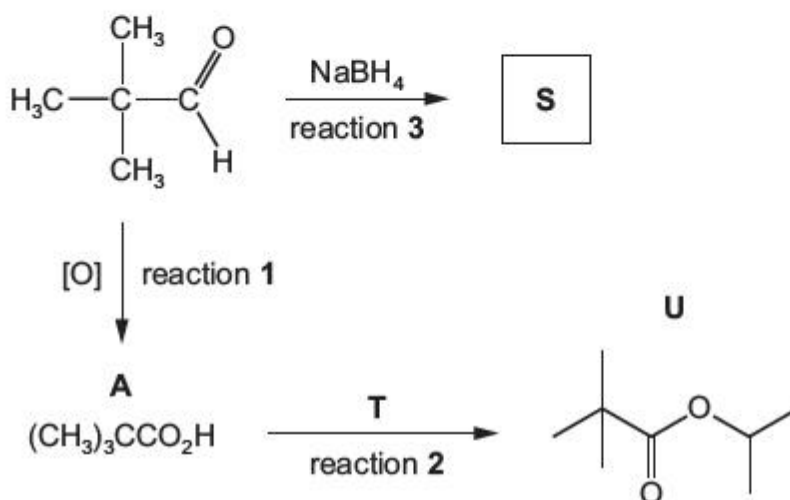
..... [1]

(ii) State the hybridisation of the carbon atom labelled with an asterisk, *.



..... [1]

(b) Two reaction sequences are shown.



- (i) Reaction 1 is an oxidation reaction.

Identify the reagent(s) and conditions for reaction 1.

..... [1]

- (iii) Give the balanced equation for the reaction of $(\text{CH}_3)_3\text{CCHO}$ with NaBH_4 to form **S**.

Use [H] to represent an atom of hydrogen provided by NaBH_4 .

..... [1]

- (c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

- (i) **X** and **Y** each contains a carbonyl group.

Complete the table with the expected observations for the reactions of **X** and **Y** with 2,4-DNPH. [1]

- (ii) Identify the functional group present in **Y** that causes the recorded observation with Fehling's solution.

..... [1]

- (iii) **Y** has a chiral centre and exists as a pair of optical isomers.

State what is meant by the term *chiral centre*.

.....

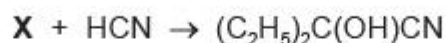
..... [1]

- (iv) Draw the optical isomers of **Y** using the conventional three-dimensional representation.

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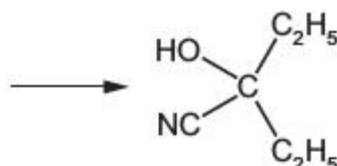
[2]

- (vii) **X** contains a carbonyl group. **X** reacts with HCN, in the presence of a small amount of NaCN, to form $(\text{C}_2\text{H}_5)_2\text{C}(\text{OH})\text{CN}$ as shown.



Draw the mechanism of the reaction of **X** with HCN.

- Draw the structure of **X** and the intermediate.
- Include all charges, partial charges, lone pairs and curly arrows.



[3]

- (viii) State the role of NaCN in the reaction in (c)(vii).

[1]

[Total: 22]

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- 5 Ethanal reacts with a mixture of HCN and NaCN to make 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

The reaction mechanism is nucleophilic addition.

- (a) Explain the meaning of the term *nucleophile* and identify the species which acts as the nucleophile during this reaction.

.....

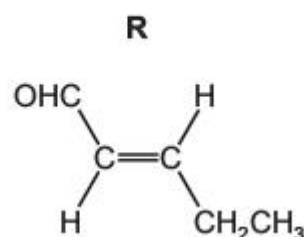
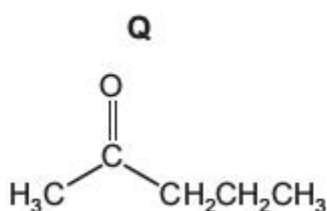
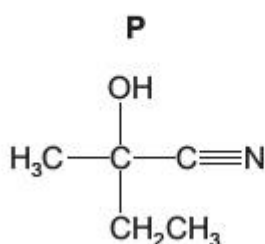
.....

species acting as nucleophile

[2]

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- 3 **P**, **Q** and **R** all contain five carbon atoms.



(b) **Q** is reduced by NaBH_4 .

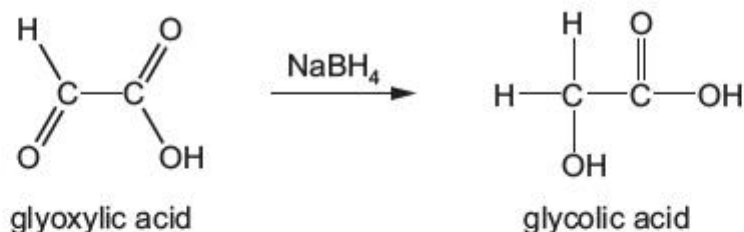
Write an equation for the reaction of **Q** with NaBH_4 .

In your answer, use $[\text{H}]$ to represent NaBH_4 .

$\text{C}_5\text{H}_{10}\text{O} + \dots\dots\dots$ [1]

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(c) Glycolic acid can also be made by reacting glyoxylic acid with NaBH_4 .



(i) State the role of NaBH_4 in this reaction.

$\dots\dots\dots$ [1]

(ii) Write an equation for this reaction using molecular formulae.
Use $[\text{H}]$ to represent NaBH_4 .

$\dots\dots\dots$ [2]

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4 **P**, **Q** and **R** all have the molecular formula $\text{C}_3\text{H}_6\text{O}$. They are all structural isomers of each other.

(c) **P** and **Q** ($\text{C}_3\text{H}_6\text{O}$) both form an orange precipitate when reacted with 2,4-DNPH. Only **Q** produces a yellow precipitate when reacted with alkaline aqueous iodine.

(i) Name **P** and **Q**.

P $\dots\dots\dots$

Q $\dots\dots\dots$

[2]

(ii) Identify the yellow precipitate formed by the reaction of **Q** with alkaline aqueous iodine.

$\dots\dots\dots$ [1]

- (ii) Ethanal, CH_3CHO , also reacts with hydrogen cyanide. The product of this reaction is $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

Draw the mechanism of this reaction.

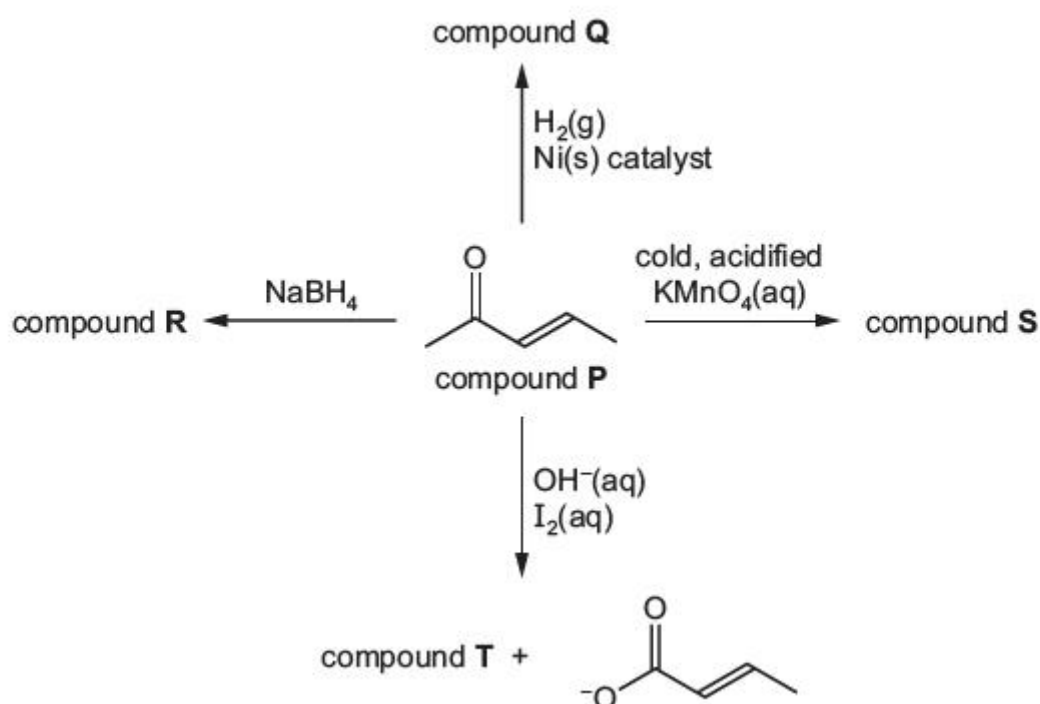
Include all necessary charges, dipoles, lone pairs and curly arrows.

[3]

[Total: 19]

Q# 385/ ALvl Chemistry/2016/m/TZ 2/Paper 4/Q# 5/www.SmashingScience.org

- 5 Some reactions of compound **P**, $\text{C}_5\text{H}_8\text{O}$, are shown.



- (ii) Give the systematic name of compound **P**.

..... [1]

- (iii) What would you observe when compound **P** is reacted with 2,4-dinitrophenylhydrazine (2,4-DNPH)?

..... [1]

(d) The reaction of hydrogen cyanide with propanone is an important first step in many organic syntheses.

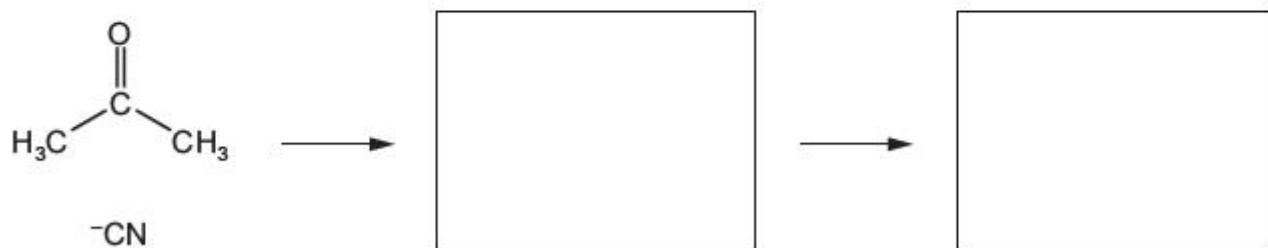
(i) Give the full name of the mechanism of this reaction.

..... [1]

(ii) Complete the diagram to show the mechanism of the reaction of hydrogen cyanide with propanone.

Draw the structure of the intermediate and the product of the reaction.

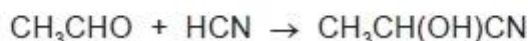
Include all relevant charges, partial charges, curly arrows and lone pairs.



[5]

[Total: 17]

3 Ethanal reacts with hydrogen cyanide, in the presence of a small amount of NaCN, as shown.

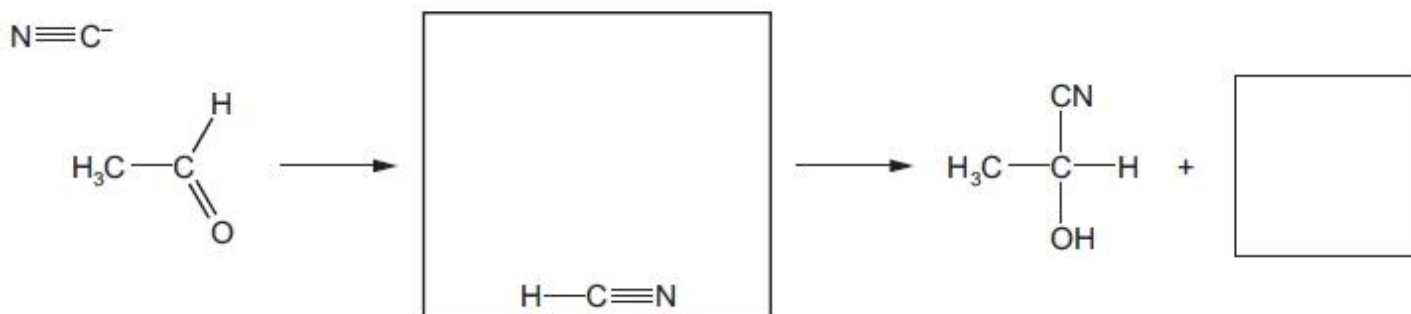


(b) The product of this reaction shows stereoisomerism as it contains a chiral centre. This reaction produces an equimolar mixture of two optical isomers.

(ii) Suggest why the two optical isomers are produced in equal amounts by this reaction.

..... [1]

(c) (i) Complete the diagram to show the mechanism of this reaction. Include all necessary charges, partial charges, lone pairs and curly arrows and show the structure of the intermediate.



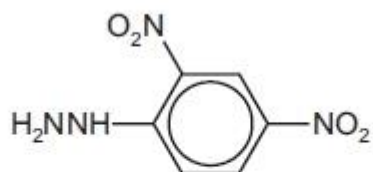
[5]

(ii) With reference to your mechanism in (i), explain the role of the NaCN in this reaction.

..... [1]

- 4 Many organic compounds, including alcohols, carbonyl compounds, carboxylic acids and esters, contain oxygen.

(b) Some oxygen-containing compounds react with 2,4-dinitrophenylhydrazine.



2,4-dinitrophenylhydrazine

- (i) Draw the structural formula of the organic compound formed when $\text{HOCH}_2\text{CH}_2\text{CHO}$ reacts with 2,4-dinitrophenylhydrazine reagent.

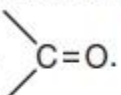
- (ii) Suggest the colour of the organic product.

.....

[2]

[Total: 12]

- 4 Ketones are widely used as solvents and as intermediates in the chemical industry.

Ketones contain the reactive keto group, .

- (a) Propanone, CH_3COCH_3 , undergoes a reaction with hydrogen cyanide, HCN .

- (i) What type of reaction is this?

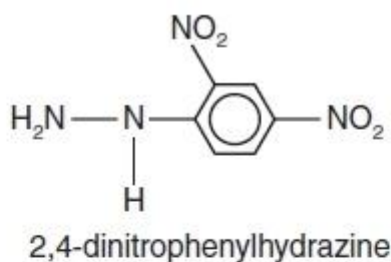
.....

- (ii) What reagents are used?

.....

- (iii) Draw a diagram to show the dipole present in the propanone molecule.

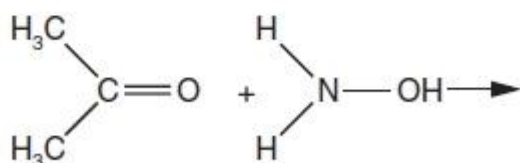
(b) Propanone reacts with 2,4-dinitrophenylhydrazine reagent.



(i) Construct a balanced equation for the reaction between propanone and 2,4-dinitrophenylhydrazine.

(ii) A similar type of reaction occurs between propanone and hydroxylamine, NH_2OH .

Draw the displayed formula of the organic product of this reaction.



[3]

[Total: 6]

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(e) HCN reacts with ethanal, CH_3CHO .

(i) Give the **displayed formula** of the organic product formed.

(ii) What type of reaction is this?

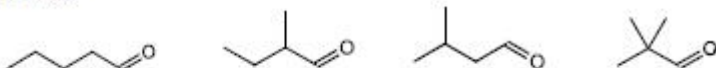
.....

- (iii) Draw the mechanism of this reaction. You should show all full and partial charges and represent the movement of electron pairs by curly arrows.

[5]

Mark Scheme ALvl Chem 17 EQ P2 22w to Q9s Paper 2 Carbonyl compounds 75marks

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4(c)(i)	carbonyl	1
4(c)(ii)	CHI ₃	1
4(c)(iii)	M1 / M2  M3 chain (isomerism)	3

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4(a)(i)	red / orange / yellow precipitate / ppt / solid [1] silver mirror / silver / grey solid / precipitate / ppt [1] effervescence / bubbling / fizzing [1]	3
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4(b)	M1: red / orange / yellow ppt / solid M2: silver mirror OR silver / grey / black / brown ppt / solid	2
4(d)(i)	CH ₂ OHCHO + 2[H] → (CH ₂ OH) ₂	1
4(d)(ii)	NaBH ₄ / LiAlH ₄	1

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4(c)(vi)	CH ₃ COCH ₃ + 3I ₂ + 4OH ⁻ → (1)CH ₃ COO ⁻ + 3H ₂ O + 3I ⁻ + CHI ₃ M1: correctly balanced M2: CHI ₃ product	2
4(c)(vii)	yellow ppt / yellow solid	1

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3(c)(i)	M1: potassium dichromate[(VI)] M2: acid(ified) AND (heat under) reflux	2
3(c)(ii)	(M1: correct identity of R and statement re: reaction 3 ONLY ketone reduced) R (is 2-hydroxybutanoic acid) AND as (only) C=O / ketone reduced (M2: correct explanation re: strength of reducing agents) NaBH ₄ cannot reduce the COOH / carboxylic acid OR LiAlH ₄ can reduce the COOH / carboxylic acid	2

3(c)(iii)	 M1: Presence of :CN (if bonding shown, must be unambiguous triple bond) M2: curly arrow from :CN lone pair to carbonyl carbon M3: correct dipole AND curly arrow from double bond to oxygen M4: correct intermediate drawn	4
3(c)(iv)	$\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{CN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$	1
3(c)(v)	Any two of three absorption references: • absorption 2200–2250 (cm^{-1}) shows presence of $\text{C}\equiv\text{N}$ • lack of absorption at 1680–1730 (cm^{-1}) shows lack of $\text{C}=\text{O}$ • lack of absorption at 2500–3000 (cm^{-1}) shows lack of RCO_2H / $\text{O}-\text{H}$ in RCO_2H	2

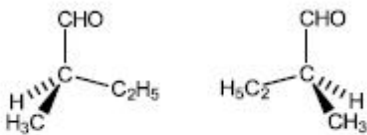
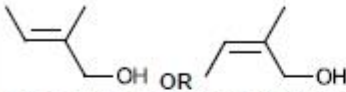
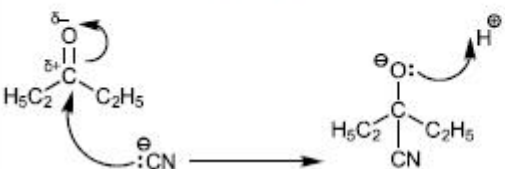
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5(c)(i)	carbonyl / aldehyde / ketone	1
5(c)(ii)	tertiary halogenoalkane	1

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3(a)(ii)	nucleophilic addition	1
3(a)(iii)	yellow / orange / red ppt / solid	1

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4(a)(i)	(2,2-)dimethylpropanal	1
4(a)(ii)	sp^2	1
4(b)(i)	acidified potassium dichromate[(VI)] AND heat under reflux	1
4(b)(iii)	$(\text{CH}_3)_3\text{CCHO} + 2[\text{H}] \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{OH}$	1
4(c)(i)	- orange / red / yellow precipitate - orange / red / yellow precipitate	1
4(c)(ii)	Aldehyde	1
4(c)(iii)	has a carbon / atom attached / bonded to four different atoms / groups / groups of atoms / chains	1
4(c)(iv)	 M1: Correct 3D representation M2: Correct 3D representation of drawn enantiomer	2
4(c)(vi)	 M1: skeletal alkene group AND C5 structure M2: one alcohol group M3: branched chain AND capable of geometrical isomerism	3
4(c)(vii)	M1: Correct structure of X and correct dipole on $\text{C}=\text{O}$ M2: curly arrow from $\text{C}=\text{O}$ bond to O AND intermediate with CN attached and -ve charge on the O M3: curly arrow from lone pair on CN^- to $\text{C}(=\text{O})$ in X AND curly arrow from lone pair in the intermediate to H^+ 	3

4(c)(viii)	catalyst	1
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5(a)	M1 a lone pair / electron pair donor	1
	M2 (:)CN ⁻ / ⁻ (:)CN / cyanide ion	1

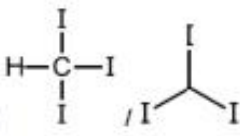
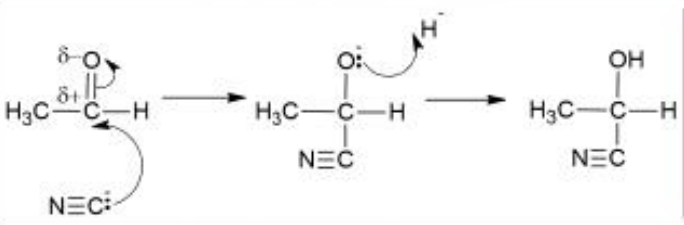
Q# 382/ ALvl Chemistry/2019/m/TZ 2/Paper 4/Q# 3/www.SmashingScience.org

3(b)	$C_5H_{10}O + 2[H] \rightarrow C_5H_{12}O$	1
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Q# 383/ ALvl Chemistry/2018/w/TZ 1/Paper 4/Q# 4/www.SmashingScience.org

4(c)(i)	reducing agent / reductant	1
4(c)(ii)	$C_2H_2O_3 + 2[H] \rightarrow C_2H_4O_3$ M1 for correct molecular formulae $C_2H_2O_3$ and $C_2H_4O_3$ M2 for balancing	2

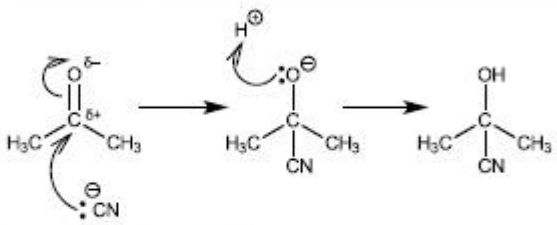
Q# 384/ ALvl Chemistry/2017/s/TZ 1/Paper 4/Q# 4/www.SmashingScience.org

4(c)(i)	P = propanal	1
	Q = propanone	1
4(c)(ii)	 tri(i)iodomethane / CHI ₃ /	1
4(d)(ii)	 curly arrow from lone pair on :C≡N to C ^(δ+) correct dipole on carbonyl $\delta^+C=\delta^-O$ AND curly arrow from bond to O ^(δ-) correct intermediate, including C-O ⁻ AND curly arrow from lone pair to H ⁺	1 1 1

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(ii)	pent-3-en(e)-2-one OR 3-penten-2-one	[1]	[1]
(iii)	red / orange / yellow precipitate / solid	[1]	[1]

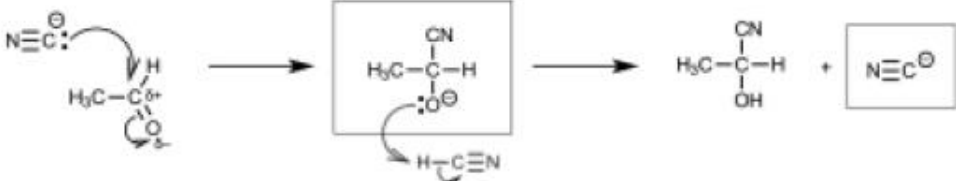
Q# 386/ ALvl Chemistry/2016/m/TZ 2/Paper 4/Q# 3/www.SmashingScience.org

(d) (i)	nucleophilic addition	[1]	[1]
(ii)	 correct dipole on carbonyl curly arrow from lone pair on CN ⁻ AND from C=O to O correct intermediate curly arrow from lone pair on O ⁻ to H ⁺ correct product	[1] [1] [1] [1] [1]	[5]

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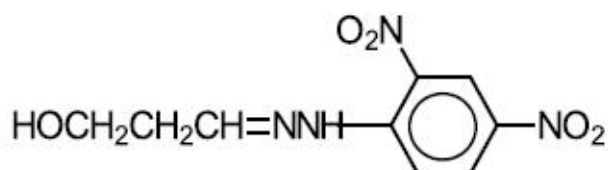
(ii)	(Planar) carbonyl so (equal chance of nucleophile) attacking either side	[1]	[1]
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3 (c) (i)	 M1 = lone pair AND curly arrow from lone pair to carbonyl C M2 = partial charges on C=O AND curly arrow from bond (=) to O^{δ-} M3 = structure of intermediate including charge M4 = lone pair AND two correct curly arrows (from lone pair to H AND from H—C to C) M5 = CN⁻	[1] [1] [1] [1]	
(ii)	(CN ⁻ regenerated so) catalyst	[1]	[5]

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(b) (i)



(1)

(ii) red or orange

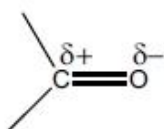
(1) [2]

Q# 389/ ALvl Chemistry/2011/s/TZ 1/Paper 4/Q# 4/www.SmashingScience.org

4 (a) (i) nucleophilic addition
both words are necessary (1)

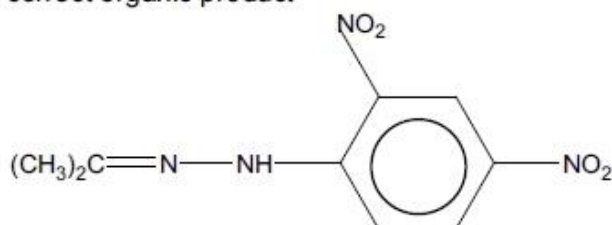
(ii) NaCN and H₂SO₄ or
HCN plus CN⁻
do not allow HCN on its own (1)

(iii) correct δ+ and δ-, i.e.



(1) [3]

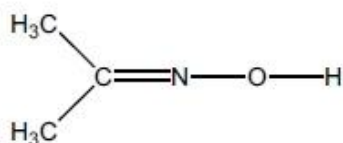
(b) (i) correct organic product



C=N bond must be clearly shown
H₂O formed/ equation balanced

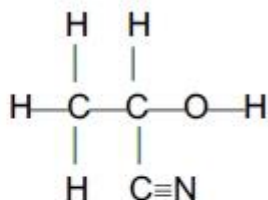
(1)
(1) [2]

(ii)



(1) [1]

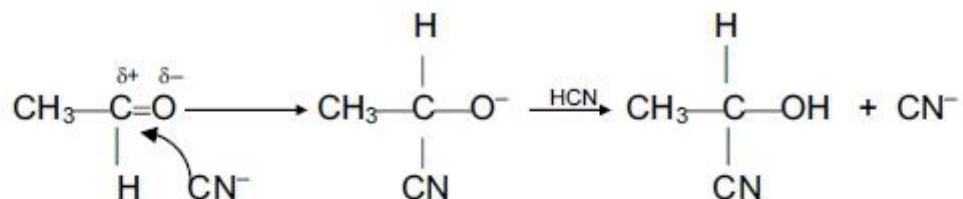
(e) (i)



C≡N must be shown (1)

(ii) nucleophilic addition (1)

(iii)



C=O dipole correctly shown or correct curly arrow on C=O (1)

attack on C^{δ+} by C of CN⁻ (1)

correct intermediate (1)

CN⁻ regenerated (1)

[5 max]

[Total: 13]

